Saeed **Boorboor**

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saeedboorboor.github.io

My research interests include designing scientific **visualization** techniques for novel domain-specific applications. Beyond the traditional desktop setup, I am also interested in **immersive** analytics of scientific data, which uses **Augmented and Virtual Reality** to support visual analysis and decision-making.

Education

2015 – 2023 **Ph.D. in Computer Science**, Stony Brook University, NY, USA.
Dissertation: Immersive and Augmented 3D Scientific Visualization Systems
Advisor: Arie E. Kaufman

2010 – 2014 **B.S. in Computer Science**, School of Science and Engineering, LUMS, Lahore, Pakistan

Academic Appointments

Aug 2025 – Now **Assistant Professor**, *Department of Computer Science*, University of Illinois, Chicago, IL, USA.
Electronic Visualization Lab

Sept 2023 – July 2025 **Principal Research Scientist**, Stony Brook University, NY, USA.
Center for Visual Computing

Aug 2016– Sept 2023 **Graduate Research Assistant**, *Center for Visual Computing*, Stony Brook University, NY, USA.
Advisor: Arie E. Kaufman

2014 – 2015 **Assistant Research Scientist**, *Technology for People Initiative*, LUMS, Pakistan.
Project: Digital Preservation of Heritage Sites

2012 – 2014 **Student Research Assistant**, *Computer Vision Lab*, LUMS, Pakistan.
Advisors: Sohaib A. Khan and Murtaza Taj

Academic Service

Conference Organization

IEEE VR, Organization Committee, Publication Chairs (2026)
IEEE VIS, Program Committee – 2024, 2025
ACM DIS, Program Committee, Artifacts and Systems, Associate Chair – 2026
IEEE VIS, Organization Committee, Supporter's Logistics – 2024
EuroVis, Program Committee – 2025
ACM IUI, Program Committee – 2025
IEEE VIS, Student Volunteer – 2018, 2022, 2023

Reviewer

Journal Transactions on Visualization and Computer Graphics, IEEE
Transactions on Emerging Topics in Computing, IEEE
Scientific Reports, Nature Publishing
Information Processing and Management, Elsevier
International Journal of Human-Computer Interaction

Conference IEEE VIS
IEEE Virtual Reality and 3D User Interface
SIGGRAPH Posters Program
IEEE Pacific Visualization Symposium
IEEE International Symposium on Mixed and Augmented Reality
ACM CHI Conference on Human Factors in Computing Systems

IEEE Eurographics / VGTC Symposium on Visualization
China Visualization and Visual Analytics Conference
Other IEEE Computer Graphics and Applications

Teaching Experience

Spring 2026 **Instructor**, *CS 427, Creative Coding*.
12 CS Undergraduate students, 15 CS+DES Undergraduate students and 4 Graduate students, University of Illinois Chicago

Fall 2025 **Instructor**, *CS 428, Virtual Reality, Augmented Reality, Mixed Reality*.
35 Undergraduate students and 5 Graduate students, University of Illinois Chicago

Spring 2024 **Instructor**, *CSE 366, Introduction to Virtual Reality*.
Undergraduate course of 35 students, Stony Brook University

Spring 2024 **Instructor**, *EMP 532, Big Data Systems for Technology Management*.
Graduate course of 11 MS and professional students, Stony Brook University

Spring 2019 – 2023 **Teaching Fellow**, *CSE 566, Virtual Reality*.
Graduate course of 25 students, Stony Brook University

Fall 2015 **Teaching Assistant**, *CSE 114, Object Oriented Programming*.
Undergraduate course of 100 students, Stony Brook University

Spring 2014 **Teaching Assistant**, *CS 200, Data Structures*.
Undergraduate course of 100 students, LUMS

Grants

Grants Awarded at Stony Brook University

2024 **Targeted Research Opportunity Program: VDR-Fusion Award: AI-Enhanced Multimodal Imaging for the Prevention of Retained Surgical Items***PIs Georgios Georgakis, Todd Griffin, Saeed Boorboor, Manoj Mahajan, and Rong Zhao.*I led the writing for the digital twin visualization subsection of this grant proposal.Award amount: $40,000 for 2 years.

Grant Writing Experience as a PhD Student

2024 **ONR N000142312124, Developing a Vegetation Management Toolkit by Integrating Multispectral/ Hyperspectral Imaging, Weather/Environmental Intelligence, and Digital Twin Visualization***PIs Arie E. Kaufman and Rong Zhao.*I led the writing for the digital twin visualization subsection of this grant proposal.

2023 **NSF 2107328-IIS, Situated Visual Information Spaces***PIs Hanspeter Pfister, Arie E. Kaufman, and James Tompkin.*I led the writing for the visualization subsection of this grant proposal.

2020 **NSF 1940302-ICER, A Coastal Alliance Network for Visualization, Assessment, Science, and Stakeholders for Convergent Environmental Problem-Solving.***PIs Brian A. Colle, Terri M. Adams-Fuller, Arie E. Kaufman, and Laura Lindenfeld.*I led the writing of the immersive visualization section of this grant proposal.

Publications

Peer-reviewed Journal Papers

2025 J[10] Yoonsang Kim, Zainab Aamir, Mithilesh Singh, **Saeed Boorboor**, Klaus Mueller, and Arie E Kaufman. Explainable XR: Understanding user behaviors of xr environments using LLM-assisted analytics framework. *IEEE Transactions on Visualization and Computer Graphics*, 2025

2023 [J9] **Saeed, Boorboor**, Matthew S Castellana, Yoonsang Kim, Chen Zhu-Tian, Johanna Beyer, Hanspeter Pfister, and Arie E Kaufman. VoxAR: Adaptive visualization of volume rendered objects in optical see-through augmented reality. *IEEE Transactions on Visualization and Computer Graphics*, 30(10):6801–6812, 2023

2023 [J8] **Saeed, Boorboor**, Yoonsang Kim, Ping Hu, Josef M Moses, Brian A Colle, and Arie E Kaufman. Submerse: Visualizing storm surge flooding simulations in immersive display ecologies. *IEEE Transactions on Visualization and Computer Graphics*, 30(9):6365–6377, 2023

2023 [J7] Ping Hu, **Saeed Boorboor**, Joseph Marino, and Arie E Kaufman. Geometry-Aware Planar Embedding of Treelike Structures. *IEEE Transactions on Visualization and Computer Graphics*, 29(7):3182–3194, 2023

2023 [J6] **Saeed Boorboor**, Shawn Mathew, Mala Ananth, David Talmage, Lorna W. Role, and Arie E. Kaufman. NeuRegenerate: A Framework for Visualizing Neurodegeneration. *IEEE Transactions on Visualization and Computer Graphics*, 29(3):1625–1637, 2023

2023 [J5] Brian A Colle, Julia R Hathaway, Elizabeth J Bojsza, Josef M Moses, Shadya J Sanders, Katherine E Rowan, Abigail L Hils, Elizabeth C Duesterhoeft, **Saeed, Boorboor**, Arie E Kaufman, and Susan E Brennan. Risk perception and preparation for storm surge flooding: A virtual workshop with visualization and stakeholder interaction. *Bulletin of the American Meteorological Society*, 2023

2022 [J4] Johanna Beyer, Jakob Troidl, **Saeed Boorboor**, Markus Hadwiger, Arie Kaufman, and Hanspeter Pfister. A Survey of Visualization and Analysis in High-Resolution Connectomics. *Computer Graphics Forum*, 41(3):573–607, 2022

2021 [J3] Parmida Ghahremani, **Saeed Boorboor**, Pooya Mirhosseini, Chetan Gudisagar, Mala Ananth, David Talmage, Lorna W Role, and Arie E Kaufman. Neuroconstruct: 3D reconstruction and visualization of neurites in optical microscopy brain images. *IEEE Transactions on Visualization and Computer Graphics*, 28(12):4951–4965, 2021

2019 [J2] Ji Hwan Park, Saad Nadeem, **Saeed Boorboor**, Joseph Marino, and Arie Kaufman. CMed: Crowd analytics for medical imaging data. *IEEE Transactions on Visualization and Computer Graphics*, 27(6):2869–2880, 2019

2018 [J1] **Saeed Boorboor**, Shreeraj Jadhav, Mala Ananth, David Talmage, Lorna Role, and Arie Kaufman. Visualization of neuronal structures in wide-field microscopy brain images. *IEEE Transactions on Visualization and Computer Graphics*, 25(1):1018–1028, 2018

Peer-reviewed Conference Papers

2025 C[8] Matthew S Castellana, Chahat Kalsi, Yoonsang Kim, **Saeed Boorboor**, and Arie E Kaufman. AuxiScope: Handheld augmented reality tablet as an auxiliary display for large-scale display systems. In *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 1660–1670, 2025

2025 C[7] **Saeed Boorboor**, Doris Gutiérrez-Rosales, Ahamed Shoaib, Chahat Kalsi, Yue Wang, Yuyang Cao, Xianfeng Gu, and Arie E Kaufman. Silo: Half-gigapixel cylindrical stereoscopic immersive display. In *IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, pages 483–493, 2025

2023 [C6] Josef Moses, Brian Anthony Colle, Julia Roberts Hathaway, Elizabeth Bojsza, Shadya Sanders, Katherine E Rowan, Abigail Hills, Elizabeth C Duesterhoeft, **Saeed, Boorboor**, Arie E Kaufman, et al. A virtual workshop involving college students to explore the relative role of visualization and stakeholder interaction on risk perception. In *Conference on Weather Analysis and Forecasting, Numerical Weather Prediction, and Mesoscale Processes*. AMS, 2023

2022 [C5] Ping Hu, **Saeed Boorboor**, Shreeraj Jadhav, Joseph Marino, Seyedkoosha Mirhosseini, and Arie E Kaufman. Spatial Perception in Immersive Visualization: A Study and Findings. In *International Symposium on Mixed and Augmented Reality Adjunct* , pages 369–372. IEEE, 2022

2021 [C4] Yoonsang Kim, **Saeed Boorboor**, Amir Rahmati, and A Kaufman. Design of privacy preservation system in augmented reality. In *IEEE Symposium on Visualization for Cyber Security*, 2021

2018 [C3] Mala Ananth, **Saeed Boorboor**, Shreeraj Jadhav, Arie Kaufman, Mark Slifstein, Nikhil Palekar, Christine DeLorenzo, Ramin Parsey, David Talmage, and Lorna Role. Visualizing the cholinergic system in health & disease. In *Neuropsychopharmacology*, volume 43, pages S263–S264. Nature Publishing Group, 2018

2018 [C2] **Saeed Boorboor**, Saad Nadeem, Ji Hwan Park, Kevin Baker, and Arie Kaufman. Crowdsourcing lung nodules detection and annotation. In *Medical Imaging 2018: Imaging Informatics for Healthcare, Research, and Applications*, volume 10579, pages 342–348. SPIE, 2018

2015 [C1] Ahsan Abdullah, Reema Bajwa, Syed Rizwan Gilani, Zuha Agha, **Saeed Boorboor**, Murtaza Taj, and Sohaib Ahmed Khan. 3D Architectural Modeling: Efficient RANSAC for n-gonal Primitive Fitting. In *Eurographics*, pages 5–8, 2015

Peer-reviewed Workshops, Posters, and Abstracts

2026 W[2] Ava Nederlander, Matthew Castellana, **Saeed Boorboor**, Arie Kaufman, and Georgios Georgakis. Abdominal Cavity Reconstruction from Laparoscopic PIPAC Video: Toward Quantitative and Qualitative Assessment of Peritoneal Surface Disease. *Society of Surgical Oncology (SSO), 20th Advanced Cancer Therapies, 2025. Selected for Oral Presentation.*

2025 W[1] Chahat Kalsi, Yuancheng Shen, Sophia Gaupp, and Luca Reichmann, Meri Rogava, Michael Krone, **Saeed Boorboor**, and Robert Krüger. BioSET - Biomarker-based Spatial co-Expression analysis in Tumor environments. *In IEEE BioMedVis Challenge 2025, IEEE VIS, 2025.*

Patents and Licenses

P1 System, Method, and Computer-Accessible Medium for Processing Brain Images and Extracting Neuronal Structures. A Kaufman and **S Boorboor**. US Patent App. 16/994,885.

P2 Method and System for Color Rendering in Augmented and Mixed Reality. A Kaufman and **S Boorboor**. Copyrighted software to facilitate the visualization of virtual objects from their environment in augmented and mixed reality. Technology ID: 050-9419.

Mentoring Experience

Doctoral Students

2026 – Now **Hossein Fathollahian**, *University of Illinois Chicago*.

2025 – Now **Shreyas Kulkarni**, *University of Illinois Chicago*.

2023 – Now **Zainab Aamir**, *Stony Brook University*, [J10].

2022 – Now **Doris Gutierrez**, *Stony Brook University*, [C7].

2021 – Now **Yoonsang Kim**, *Stony Brook Univesity*, [J8, J9, J10, C1].

2021 – Now **Matthew Castellana**, *Stony Brook University*, [J8, J9, C8].

Masters Students

2025 – Now **Riccardo Bonfanti**, *University of Illinois Chicago*.

2024 – 2025 **Chahat Kalsi**, *Stony Brook University*, [C7, C8].

2024 – 2025 **Ahamed Shoaib**, *Stony Brook University*, [C7].

Other

Spring 2024 Mentored 3 undergraduate students for CSE487: Research in Computer Science.
Stony Brook University.

2020 - Now Supervision of undergraduate senior-year project. *LUMS, Pakistan.*
Mentored 10 undergraduate students.

Summer 2020 Supervised 2 high school summer research interns.

Awards and Honors

2024 Recognition by the National Academy of Inventors - Chapter for Inventions at Stony Brook University.

2019, 2021 IACS Junior Researcher Award. Awarded for strong contributions in computational research.
($37,000 + $5000 travel grant.)

2016 Stony Brook University CS Department Chair Fellowship: Awarded the special chair fellowship for the top 10 Ph.D. applications. ($10,000)

Selected Talks and Presentations

2025 Invited Lecture on Situated Visualization, *New York University, USA.*

2024 Visual Analysis of Volumetric Videos, *Harvard University, USA.*

2022 Visualization in Immersive Display Ecologies, *Center for Visual and Decision Informatics*.

2018 Visualization of Neuronal Structures in Optical Microscopy, *Harvard University, USA.*

2018 Visualization for Neuroscience and Connectomics. *LUMS, Pakistan.*